



S1.1 Introduction to the particulate nature of matter

Structure 1 · Models of the particulate nature of matter · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Matter is composed of particles (atoms, molecules or ions) in constant motion, and the kinetic molecular theory explains the properties and interconversion of the solid, liquid and gas states in terms of particle spacing, arrangement and energy. Pure substances are elements (one type of atom) or compounds (two or more elements chemically combined in fixed ratio), while mixtures contain substances that are not chemically bonded and can be separated by physical methods such as filtration, distillation, and chromatography. Physical changes (e.g. changes of state) do not alter chemical composition, whereas chemical changes produce new substances with different properties.

Worked example: follow the reasoning

A student runs paper chromatography on a green ink using a solvent. The ink spot travels 2.4 cm and the solvent front travels 8.0 cm from the baseline.

Step 1: $R_f = \text{distance travelled by spot} \div \text{distance travelled by solvent front}$.

Step 2: $R_f = 2.4 / 8.0 = 0.30$.

Answer: $R_f = 0.30$ for that component of the ink.

Quick check · multiple choice

Which change is a chemical change?

- A Ice melting into water
- B Iodine sublimating on heating
- C Magnesium burning in oxygen
- D Sugar dissolving in hot tea

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A student is given a mixture of sand, salt and water. [8]
- (a) Describe a sequence of physical processes that could be used to obtain (i) dry sand and (ii) solid salt as separate pure samples from this mixture. [4]
 - (b) State one observation that would show that the sand obtained in (a) is a pure element/compound-free physical substance rather than a chemical compound of the other components. [1]
 - (c) Explain, in terms of the particulate nature of matter, why salt dissolves in water but sand does not. [3]

Final answer: _____



Answers and explanation

Question 1 [8]

(a)(i) Filter the mixture; sand is retained on the filter paper as residue, rinse and dry it. (ii) The filtrate (salt solution) is heated/evaporated (e.g. using an evaporating basin) until crystallisation occurs, or evaporated to dryness, to obtain solid salt crystals. [4 marks: filtration step, sand as residue, evaporation/crystallisation step, salt as solid obtained]

(b) Sand's properties (melting point, appearance, insolubility) are unchanged/identical before and after separation, showing no chemical reaction occurred — it was only physically mixed. [1]

(c) Salt (an ionic compound) dissociates into Na^+ and Cl^- ions which are attracted to and surrounded (hydrated) by polar water molecules, allowing separation of the lattice into solution; sand (SiO_2) is a giant covalent lattice with strong covalent bonds throughout that are not disrupted by interactions with water molecules, so its particles cannot separate and disperse. [3]

Quick check answer

C. Burning forms new substances (MgO) with different properties; the others are physical changes with no new substance formed.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.